

ECTE333

Lecture 1 – Introduction to AVR Microcontrollers

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Lecture 1's sequence

1.1 Overview of ATmega16



1.2 C Development Environment for the AVR

ECTE333's schedule

Lecture (2h)	Tutorial (1h)	Lab (2h)
➔ L1: Introduction to AVR Microcontrollers		
L2: C Programming, Digital IO	Tutorial 1	Lab 1
L3: Serial Communication		
	Tutorial 2	Lab 2
L4: Interrupts, Timers		
	Tutorial 3	Lab 3
L5: Pulse Width Modulators		
	Tutorial 4	Lab 4
L6: Analogue-to-Digital Converters		
	Tutorial 5	Lab 5
L7: Microcontroller Applications		
		Lab 6

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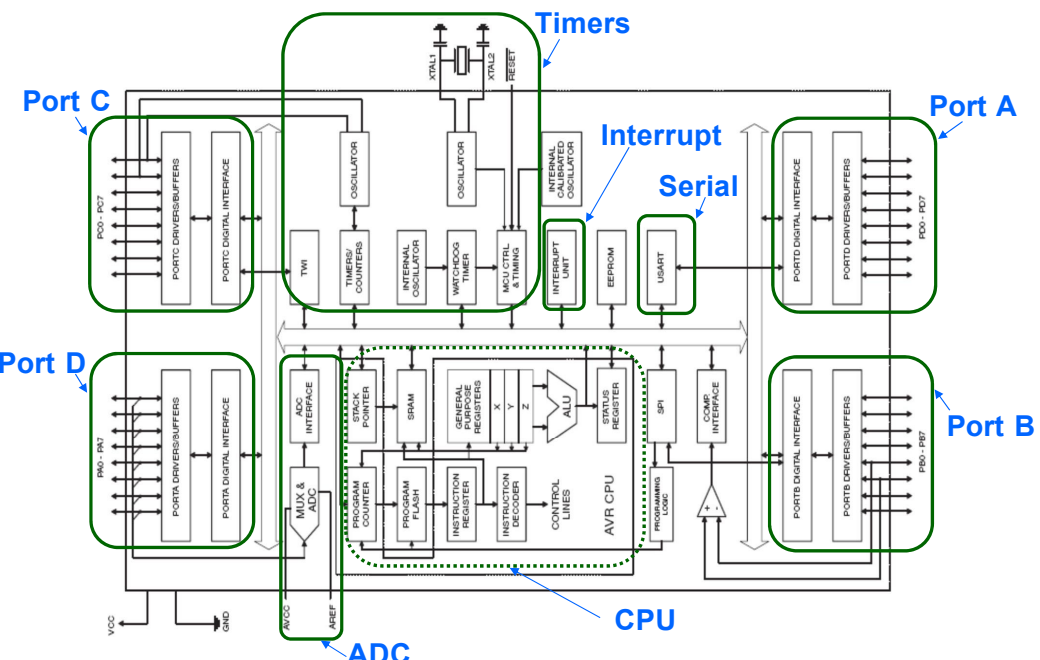
1.1 Overview of ATmega16

- To support different design requirements, the Atmel AVR family has many microcontrollers: ATmega8515, ATmega16, etc.
- Microcontrollers in the 8-bit AVR family share a similar instruction set and architecture.
- The ATmega16 has components that are useful for microcontroller applications, e.g. analogue-to-digital converter, pulse-width-modulator.

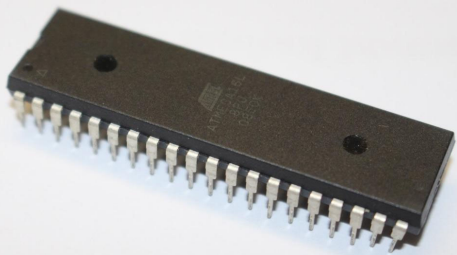
Comparison between ATmega8515 and ATmega16

	ATmega8515	ATmega16
Number of instructions	130	131
Flash program memory	8K	16K
Internal SRAM	512 bytes	1K bytes
EEPROM	512 bytes	512 bytes
External memory	up to 64K bytes	none
General-purpose 8-bit registers	32	32
IO ports	5	4
Serial interfaces	SPI, USART	SPI, USART, Two-wire
Timers	one 8-bit, one 16-bit	two 8-bit, one 16-bit
PWM channels	3	4
ADC	Analogue comparator	Analogue comparator, 8-channel 10-bit ADC
Interrupt vectors	17	21

ATmega16 – Block diagram



ATmega16 – Pin layout



ATmega16 chip

(XCK/T0) PB0	1	40	PA0 (ADC0)
(T1) PB1	2	39	PA1 (ADC1)
(INT2/AIN0) PB2	3	38	PA2 (ADC2)
(OC0/AIN1) PB3	4	37	PA3 (ADC3)
(SS) PB4	5	36	PA4 (ADC4)
(MOSI) PB5	6	35	PA5 (ADC5)
(MISO) PB6	7	34	PA6 (ADC6)
(SCK) PB7	8	33	PA7 (ADC7)
RESET	9	32	AREF
VCC	10	31	GND
GND	11	30	AVCC
XTAL2	12	29	PC7 (TOSC2)
XTAL1	13	28	PC6 (TOSC1)
(RXD) PD0	14	27	PC5 (TDI)
(TXD) PD1	15	26	PC4 (TDO)
(INT0) PD2	16	25	PC3 (TMS)
(INT1) PD3	17	24	PC2 (TCK)
(OC1B) PD4	18	23	PC1 (SDA)
(OC1A) PD5	19	22	PC0 (SCL)
(ICP1) PD6	20	21	PD7 (OC2)

Using the ATmega16 for embedded applications

Aspect	Lecture	Lab	Tutorial
Digital IO	1, 2	1	1
Serial communication	3	2	2
Interrupts, Timers	4	3	3
PWM	5	4	4
Analogue to digital converter	6	5	5
Embedded applications	7	6	-

In Lecture 2, we study digital IO ports of the ATmega16.
In Lectures 3 to 6, we'll study other peripherals.

Lecture 1's sequence

1.1 Overview of ATmega16

1.2 C Development Environment for the AVR

Top-10 Programming Languages

Rank	Language	Type	Score
1	Python	Web, Desktop, Mobile	100.0
2	Java	Web, Desktop, Mobile	95.4
3	C	Desktop, Mobile	94.7
4	C++	Desktop, Mobile	92.4
5	JavaScript	Web	88.1
6	C#	Web, Desktop, Mobile	82.4
7	R	Desktop	81.7
8	Go	Web, Desktop	77.7
9	HTML	Web	75.4
10	Swift	Mobile	70.4

Source: <https://spectrum.ieee.org/top-programming-languages>

1.2 C Development Environment for the AVR

Why using C?

- **C is a high-level programming language:** C code is easier to understand compared to other languages.
- **C supports low-level programming:** We can use C to access all hardware components of the microcontroller.
- **C has standard libraries for complex tasks:** data type conversions, standard input/output, long-integer arithmetic.
- **The Atmel AVR instruction set is designed to support C compilers:** C code is converted efficiently to assembly code.

Atmel Studio 7

- We will use Atmel Studio 7 (aka Microchip Studio 7).
- Atmel Studio is an integrated development environment for Atmel AVR microcontrollers.
- It includes text editor, C compiler, assembler, emulator, HEX file downloader.
- It is available for free at Microchip website.



Installing Atmel Studio 7

In ECTE333, we use: Atmel Studio 7 (Build 1931)

1) Uninstall earlier versions of Atmel Studio 7.

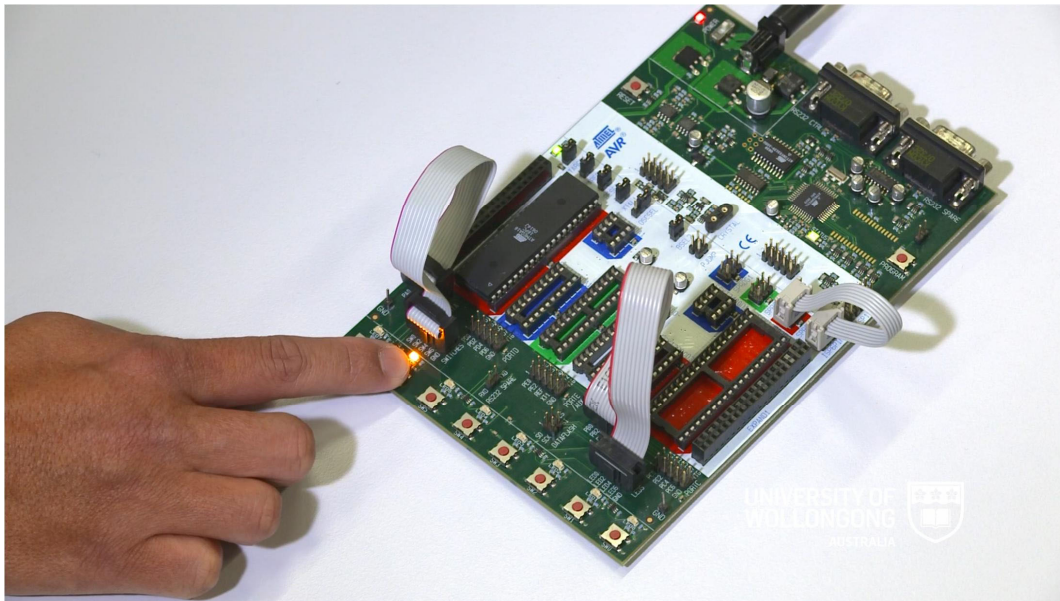
2) Download setup files for Atmel Studio 7:

www.elec.uow.edu.au/avr/tools/as-installer-7.0.1931-full.exe

3) Run setup file for Atmel Studio 7. Accept default options.

Depending on the PC, Windows Updates may be required before Step 3.

Development cycle for C – Example



Turning on LED by pressing a switch ([Lab 1, Tasks 1, 2, 3](#)).

Development cycle for C

Step 1: Creating Atmel Studio project

项目创建

- ☐ project name
- ☐ project type C
- ☐ select device

Step 2: Entering a C program.

输入C程序

Step 3: Compiling the C program to produce a HEX file.

编译C产生一个16进制文件

Step 4: Downloading/testing the HEX file on Atmel AVR microcontroller.

在单片机上测试十六进制文件

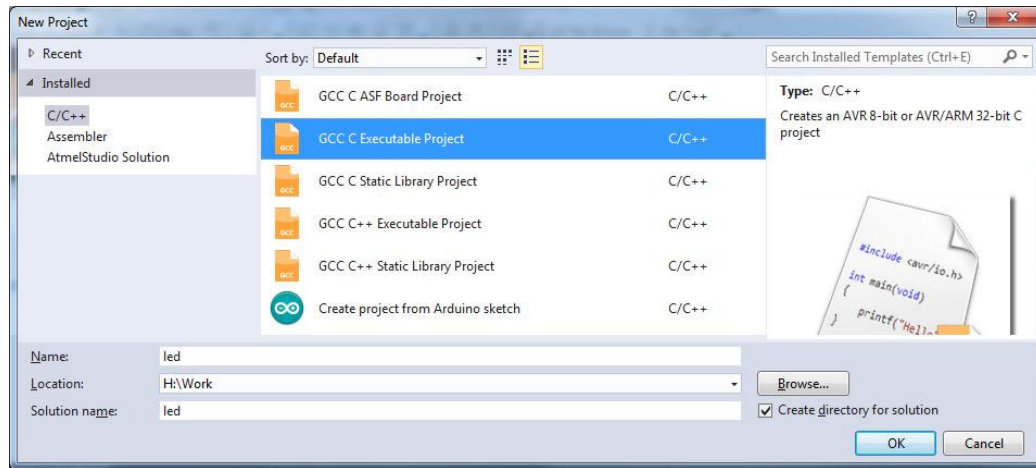
We'll illustrate these steps using an example.

Refer to the lab notes for debugging techniques.

Step 1: Creating Atmel Studio project

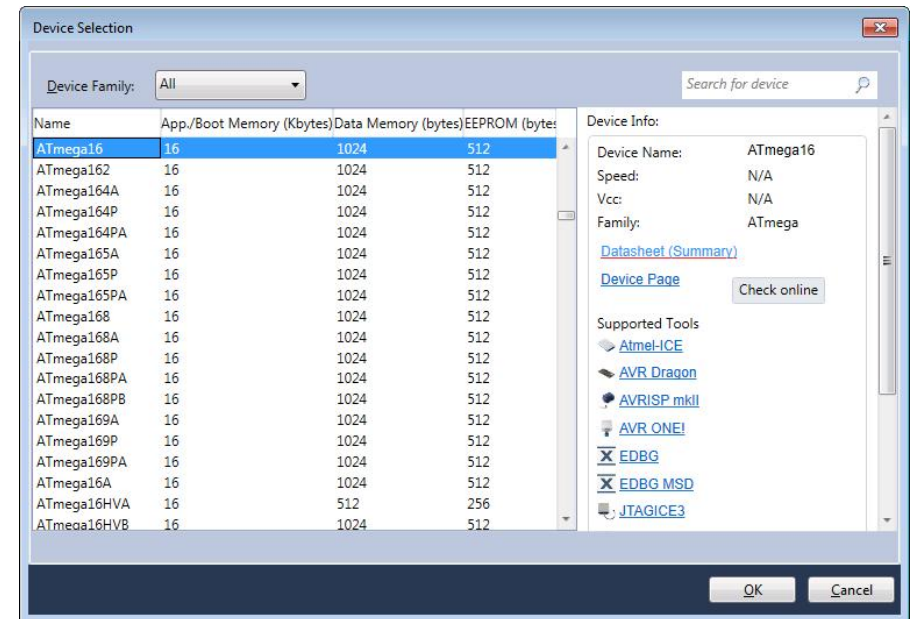
- Start the Atmel Studio 7.
- Select menu **File** | **New Project...**
 - ☐ project type: **GCC C Executable Project C/C++**
 - ☐ project name: **led**
 - ☐ project location: **H:\Work**
 - ☐ option **'Create directory for solution'**
- Click 'OK'.

Creating Atmel Studio project



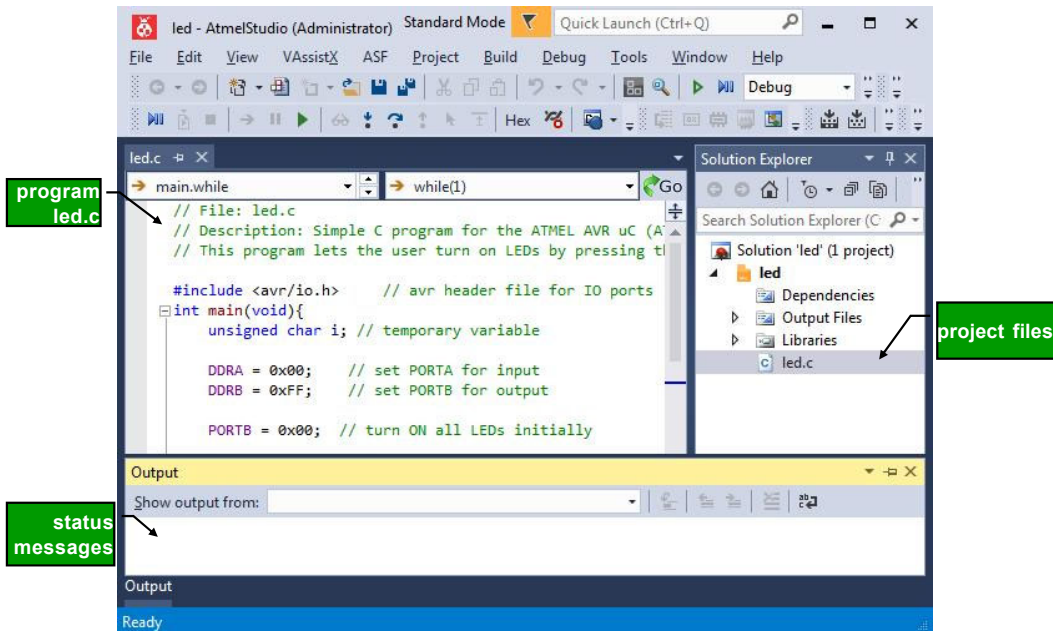
Specify project name, location, and type.

Selecting device



Search for device **ATmega16**, and click 'OK'.

Step 2: Entering a C program



The Atmel Studio window.

Entering a C program

```
/* File: led.c
Description: Simple C program for the ATMEL AVR uC (ATmega16 chip)
It lets user turn on LEDs by pressing the switches on STK500 board */

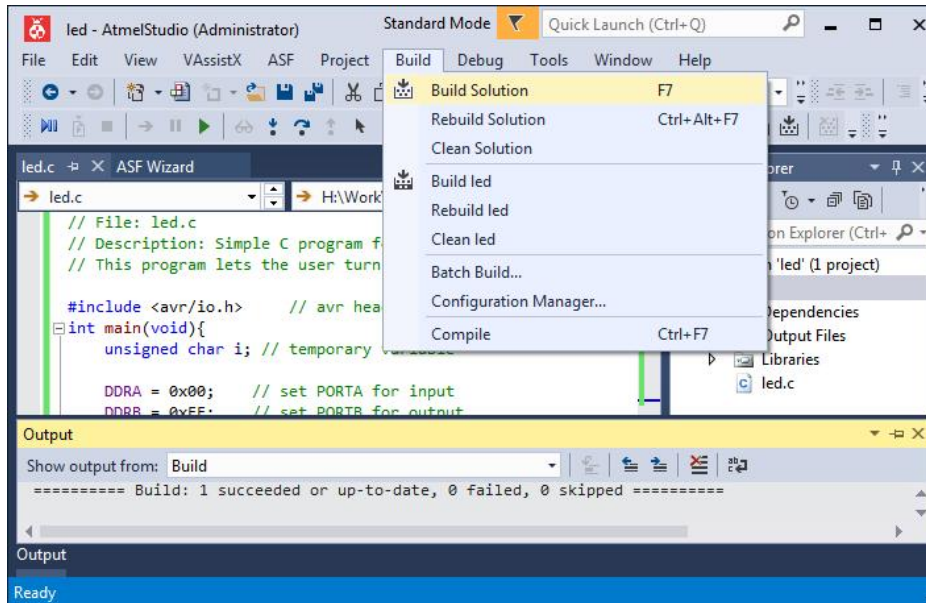
#include <avr/io.h> // AVR header file for all registers/pins
int main(void){
    unsigned char i; // temporary variable

    DDRA = 0x00; // set PORTA for input
    DDRB = 0xFF; // set PORTB for output
    PORTB = 0x00; // turn ON all LEDs initially

    while(1){
        // Read input from PORTA.
        // This port will be connected to the 8 switches
        i = PINA;

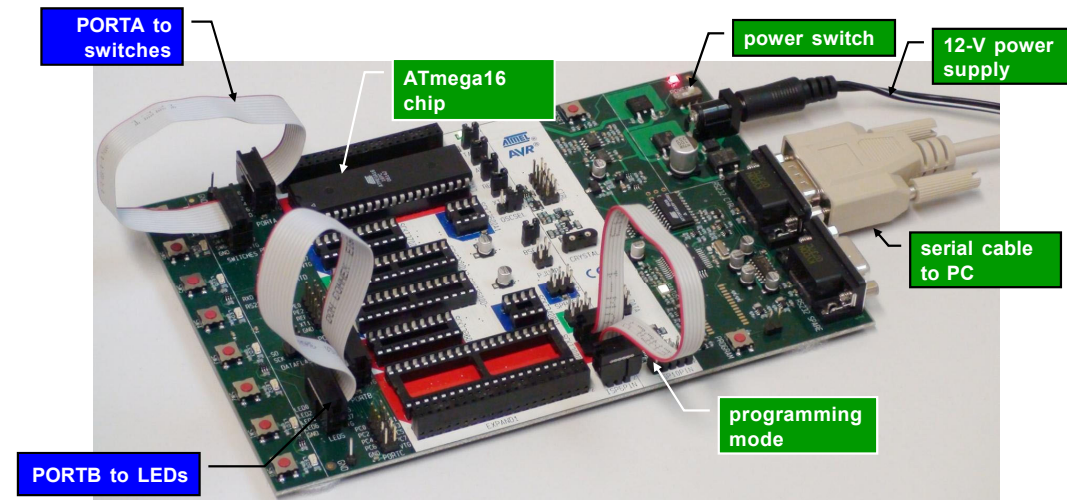
        // Send output to PORTB.
        // This port will be connected to the 8 LEDs
        PORTB = i;
    }
    return 1;
}
```

Step 3: Compiling the C program



Select menu **Build** | **Build Solution** to generate HEX file led.hex

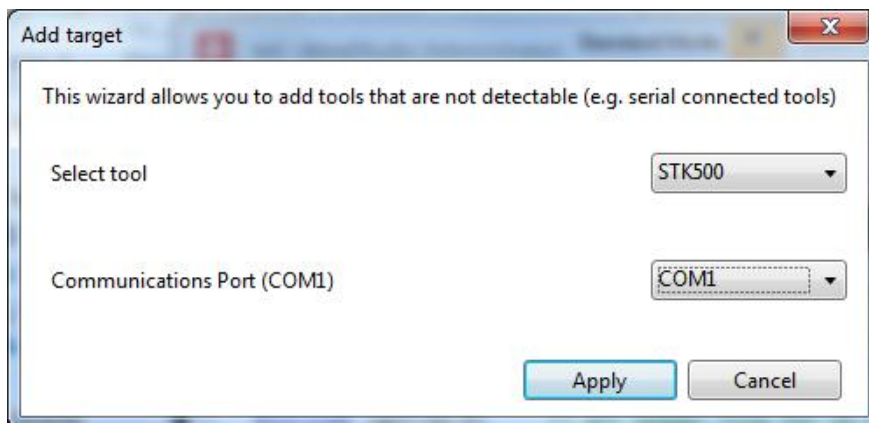
Step 4: Downloading/testing on microcontroller



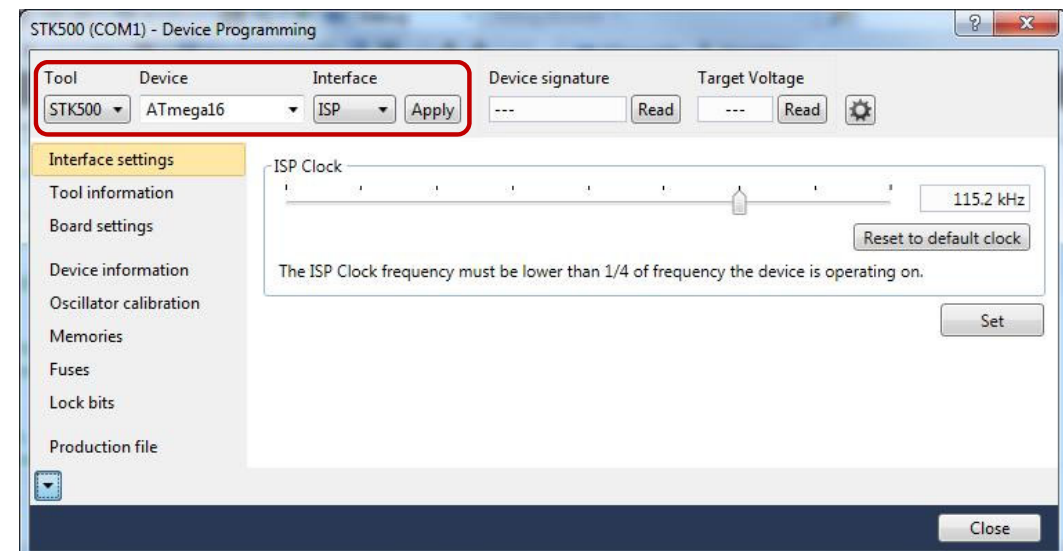
Hardware setup for example program.
Connections to PORTA & PORTB are only for this example.

Downloading/testing on microcontroller

- Select menu **Tools** | **Add target**.
- Select the correct serial port.
- Click 'Apply'.

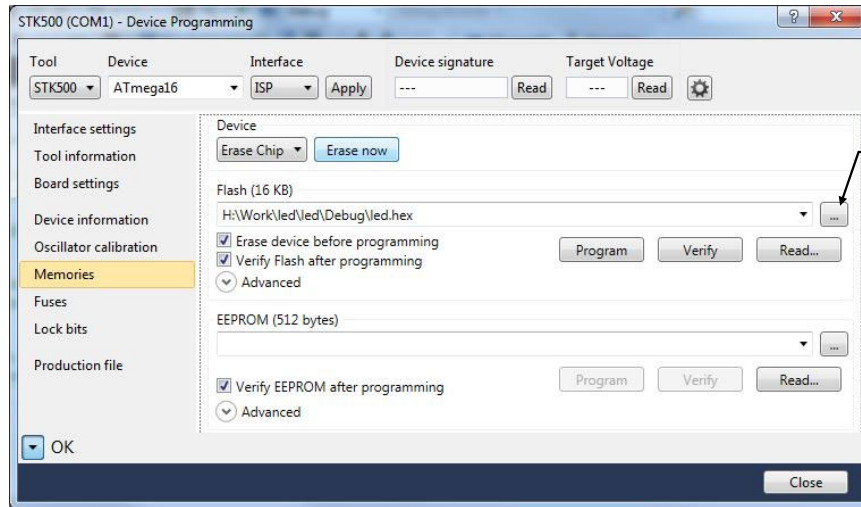


Downloading/testing on microcontroller



Select '**Tool**' = STK500, '**Device**' = ATmega16, and '**Interface**'= ISP

Downloading/testing on microcontroller



Select HEX file

- In '**Memories**' tab, select HEX file (created in Step 3) & click '**Program**'.
- After this, the program is run on the STK500 board.
- The program remains even after power-off. To erase, click '**Erase Now**'.

Lecture 1's summary

- Key points of this lecture
 - ECTE333 subject organisation.
 - An overview of ATmega16.
 - The tools and the steps for programming the Atmel AVR.
- Next activities
 - Read in advance Lecture 2.
 - Tutorial 1: 'C Programming for the AVR'.
 - Lab 1: 'C Programming for the AVR'
 - ❖ Study video demos of Lab 1.
 - ❖ Complete Tasks 1, 2, 3.
 - ❖ Complete online Pre-lab Quiz for Lab 1.
 - ❖ Write programs for Tasks 4 and 5.

Atmel AVR demo: Line following robot



Using ATmega16

Lecture 1's references

- M. A. Mazidi, S. Naimi, S. Naimi, AVR Microcontroller and Embedded Systems: Using Assembly and C, 2nd edition Pearson, 2015, [**Chapter 7: AVR Programming in C, Chapter 4: AVR I/O Port Programming**].
- J. Pardue, C Programming for Microcontrollers, 2005, SmileyMicros, [**Chapters 1-6**].
- S. F. Barrett and D. J. Pack, Atmel AVR Microcontroller Primer: Programming and Interfacing, 2008, Morgan & Claypool Publishers, [**Chapter 1**].
- D. V. Gadre, Programming and Customizing the AVR Microcontroller, 2001, McGraw-Hill, [**Chapters 1-3**].
- Atmel Corp., 8-bit AVR microcontroller with 16K Bytes In-System Programmable Flash ATmega16/ATmega16L, 2010. **Manual**

Acknowledgements

- We acknowledge the supports from **Microchip Corporation** in providing Atmel AVR equipment for teaching ECTE333 at UOW.
- The University of Wollongong is a member of the **Microchip University Program**.
- This project is supported by an ESDF teaching grant from the University of Wollongong.
- Other material for this course is available at:
www.elec.uow.edu.au/avr